Algorithms: Greedy Method

Activity-Selection Problem

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Greedy Algorithms: Principles

- A *greedy algorithm* always makes the choice that looks best at the moment.
- A greedy algorithm works in phases.

At each phase:

- You take the best you can get right now, without regard for future consequences.
- You hope that by choosing a local optimum at each step, you will end up at a global optimum.
- For some problems, it works.

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The Activity Selection Problem

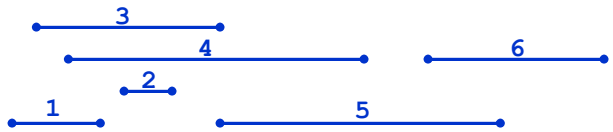
- Input: A set of activities $S = \{a_1, \ldots, a_n\}$
  - Each activity a_i has a start time s_i and a finish time f_i, where $0 \le s_i < f_i < \infty$
  - If selected, activity a_i takes place during the half-open time interval $[s_i, f_i)$
- Two activities are compatible if and only if their intervals do not overlap
- Output: a maximum-size subset of mutually compatible activities

The Activity Selection Problem

- Formally:
  - Given a set S of n activities

    s_i = start time of activity i

    f_i = finish time of activity i
  - Find max-size subset A of compatible activities



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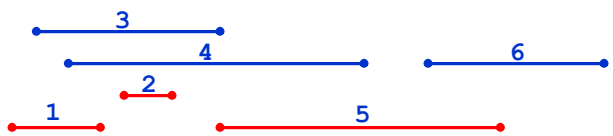

The Activity Selection Problem

- Here are a set of start and finish times

i	1	2	3	4	5	6	7	8	9	10	11
s_i	1	3	0	5	3	5	6	8	8	2	12
f_i	4	5	6	7	8	9	10	11	12	13	14

- What is the maximum number of activities that can be completed?
  - $\{a_3, a_9, a_{11}\}$ can be completed

The Activity Selection Problem

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  - But so can $\{a_1, a_4, a_8, a_{11}\}$ which is a larger set

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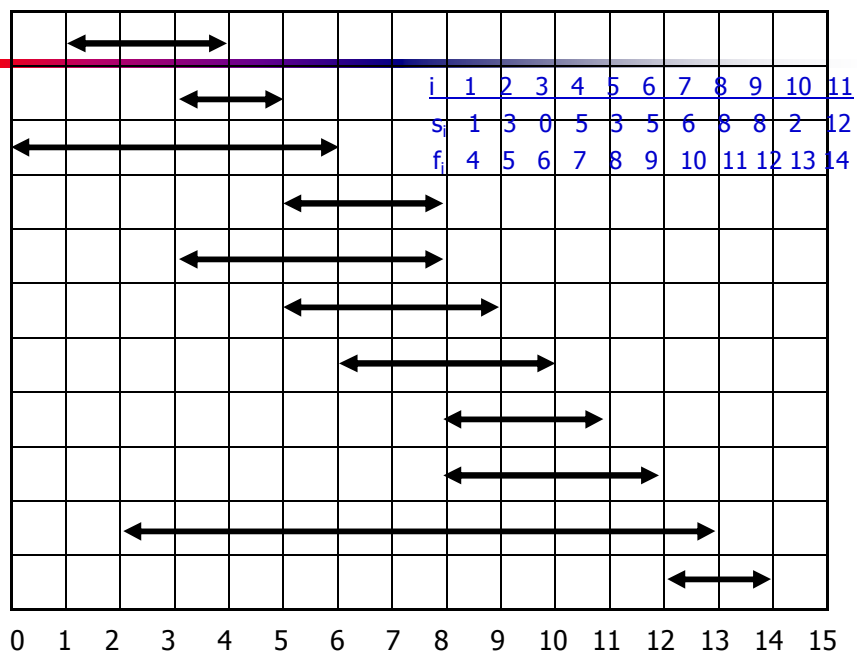
- What is the maximum number of activities that can be completed?
  - $\{a_3, a_9, a_{11}\}$ can be completed
  - But so can $\{a_1, a_4, a_8, a_{11}\}$ which is a larger set
  - But it is not unique, consider $\{a_2, a_4, a_9, a_{11}\}$

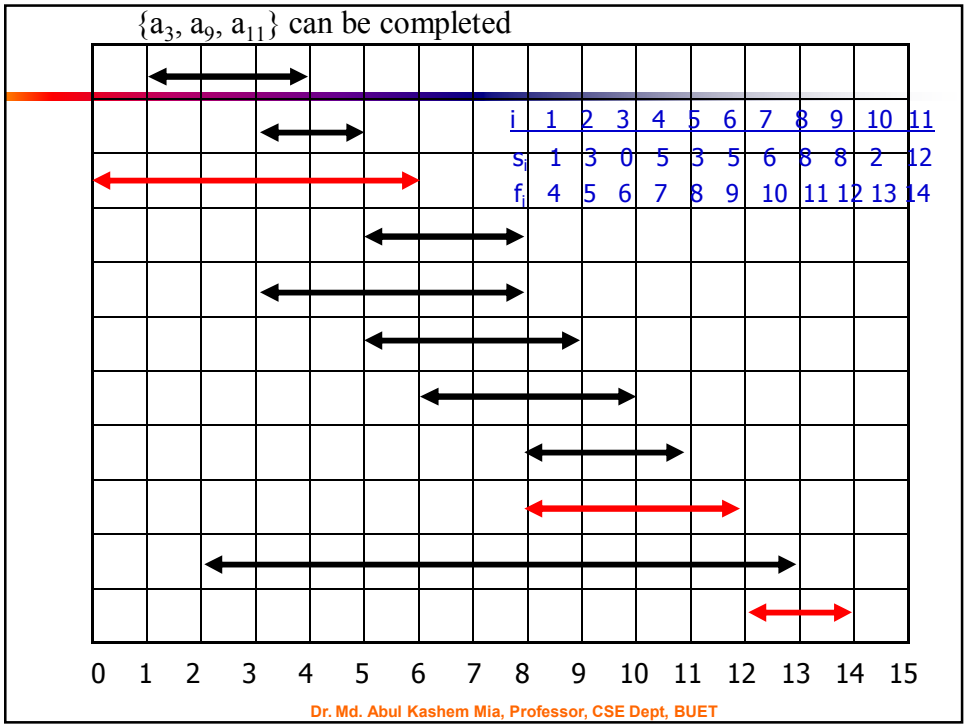
Interval Representation

- Here are a set of start and finish times

i	1	2	3	4	5	6	7	8	9	10	11
s_i	1	3	0	5	3	5	6	8	8	2	12
f_i	4	5	6	7	8	9	10	11	12	13	14



i	1	2	3	4	5	6	7	8	9	10	11
s_i	1	3	0	5	3	5	6	8	8	2	12
f_i	4	5	6	7	8	9	10	11	12	13	14


{a3, a9, a11} can be completed
i 1 2 3 4 5 6 7 8 9 10 11
si 1 3 0 5 3 5 6 8 8 2 12
fi 4 5 6 7 8 9 10 11 12 13 14
0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15
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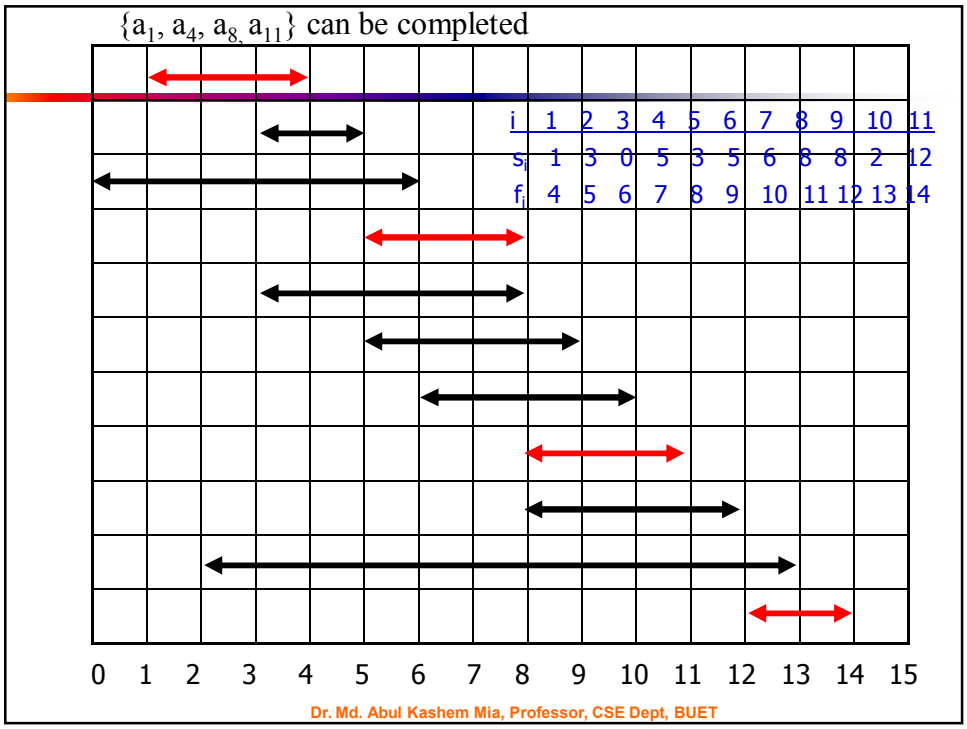
{a1, a4, a8, a11} can be completed
i 1 2 3 4 5 6 7 8 9 10 11
si 1 3 0 5 3 5 6 8 8 2 12
fi 4 5 6 7 8 9 10 11 12 13 14
0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15
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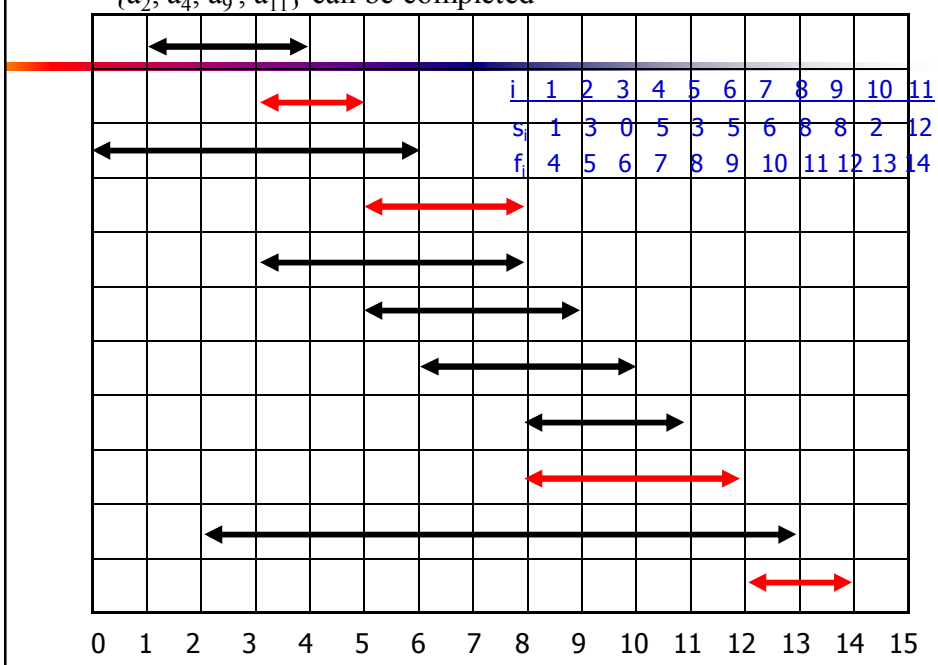
$\{a_2, a_4, a_9, a_{11}\}$ can be completed

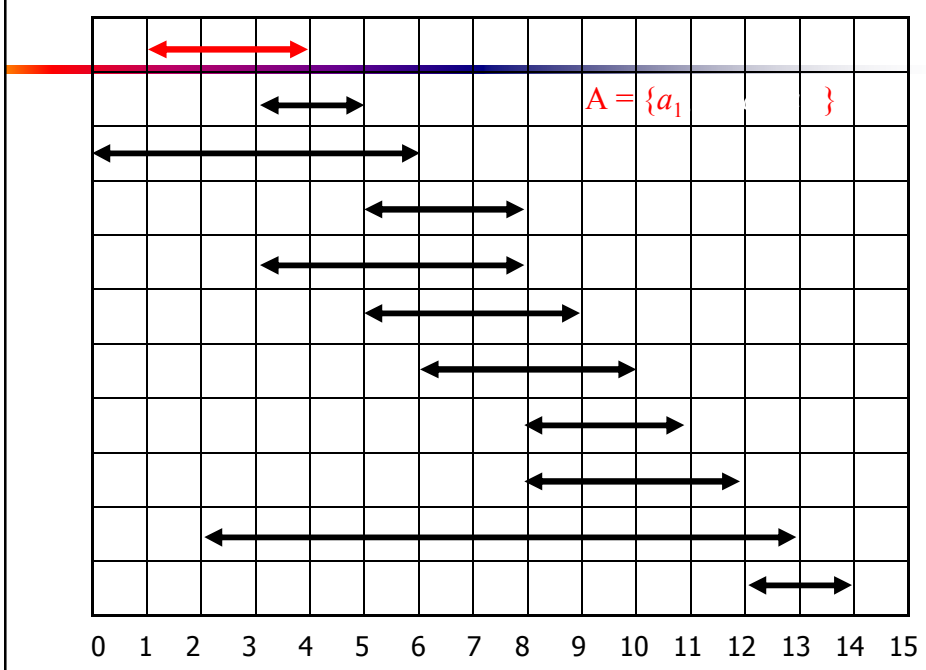


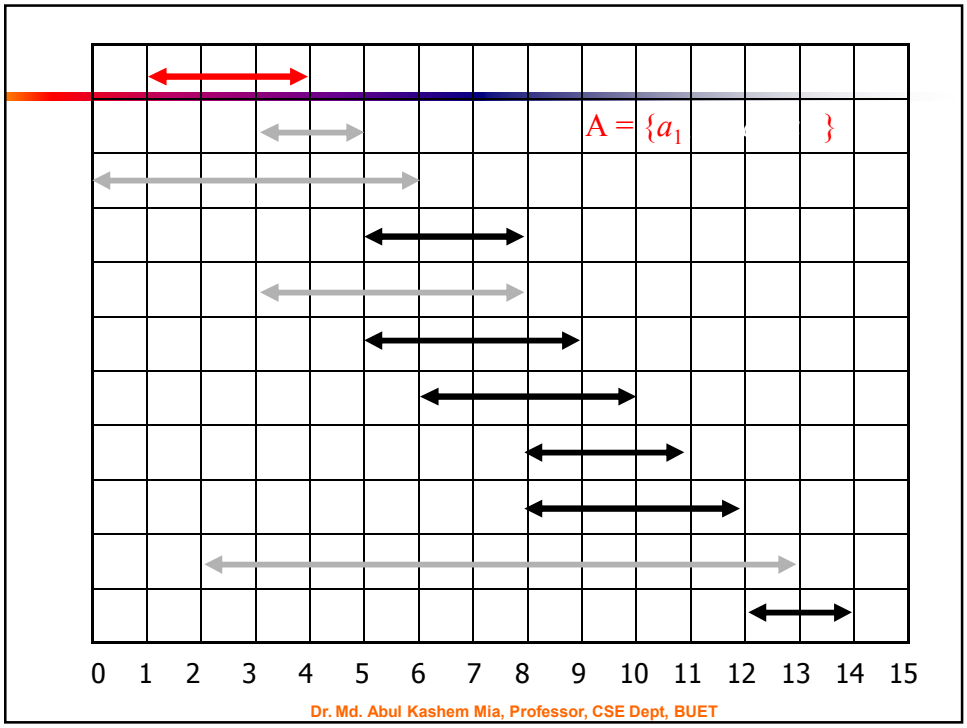
Solving the Activity Selection Problem

- The greedy choice can be applied to find *solutions* (the maximum number of activities that can be performed) by using
  - earliest finish time — Always gives optimal solution
  - earliest start time — Does not always give optimal solution
  - shortest interval — Does not always give optimal solution

Early Finish Greedy

- Select the activity with the earliest finish
- Eliminate the activities that could not be scheduled
- Repeat!




A = {a1 }
0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15
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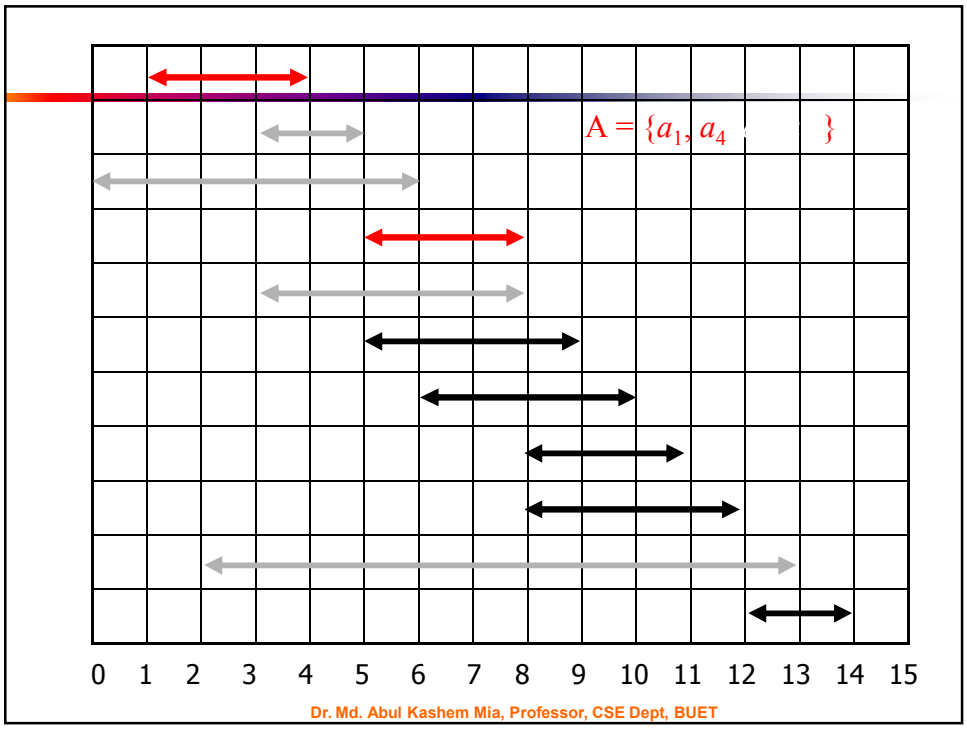
A = {a1, a4 }
0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15
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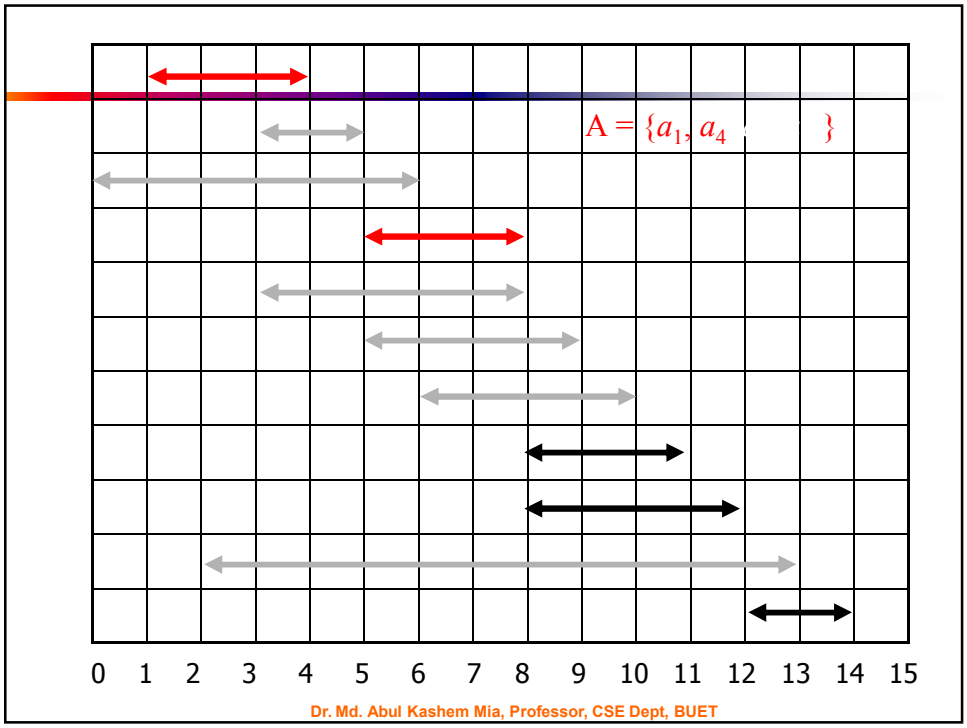
A = {a_1, a_4 }
0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15
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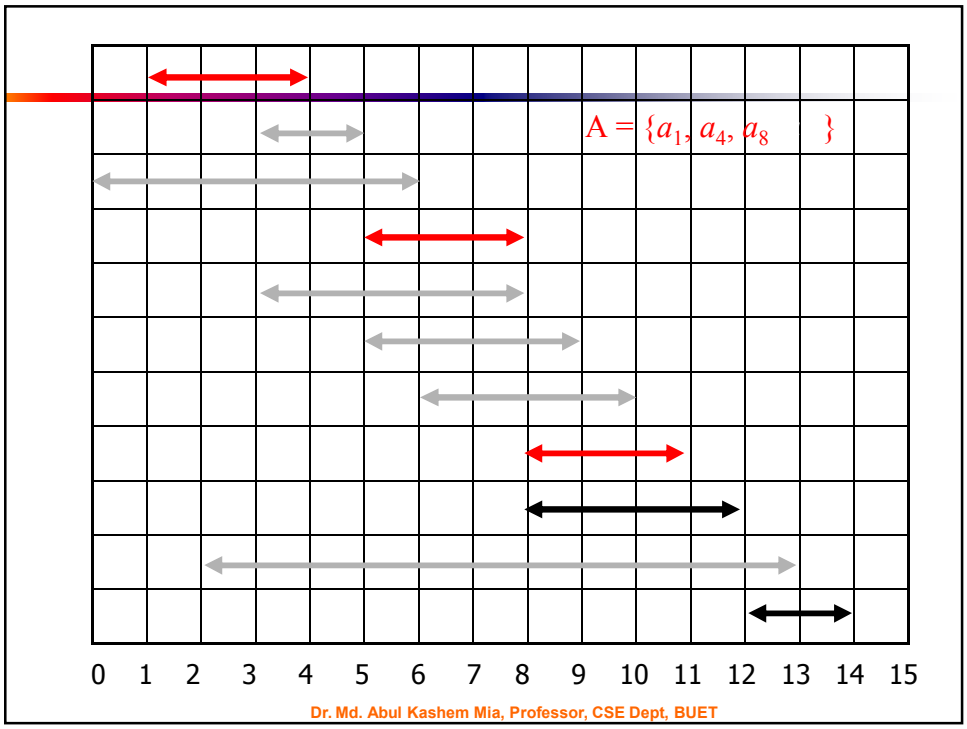
A = {a_1, a_4, a_8 }
0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15
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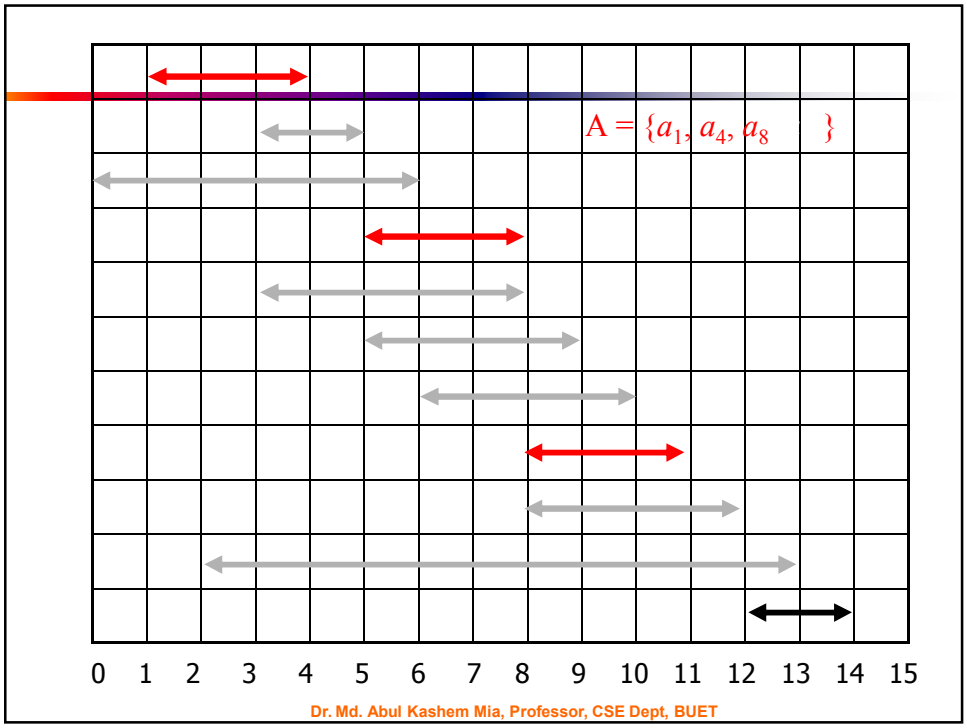
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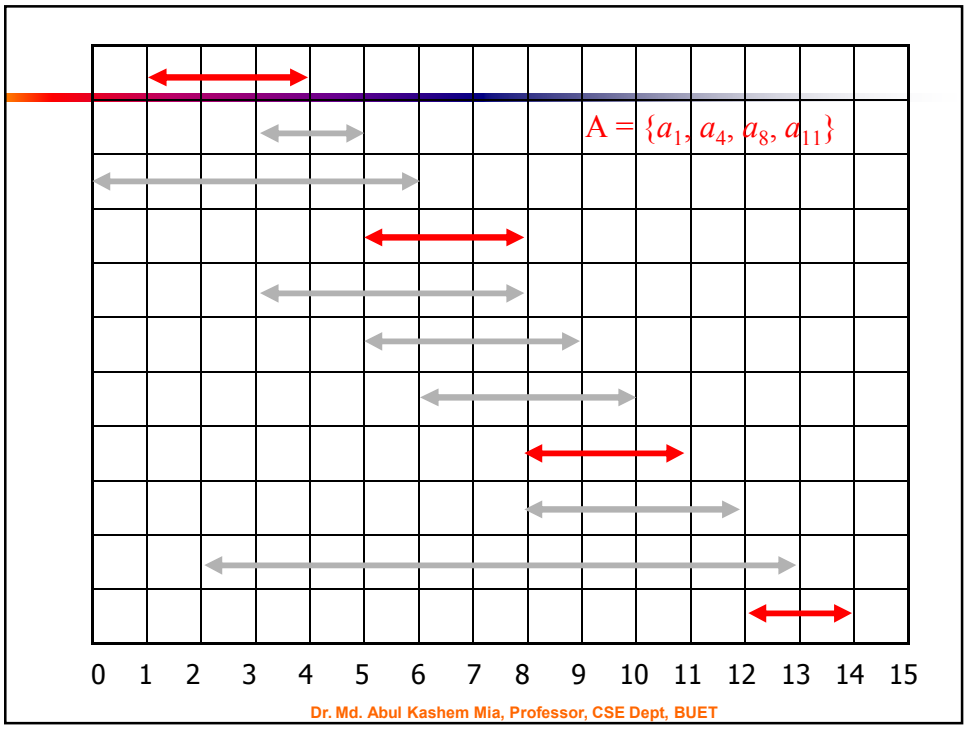
A = {a_1, a_4, a_8, a_{11}}
0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15
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Assuming activities are sorted by finish time

```
GREEDY-ACTIVITY-SELECTOR(s, f)
n ← length[s]
A ← {a_1}
i ← 1
for m ← 2 to n
      do if s_m ≥ f_i
            then A ← A ∪ {a_m}
                 i ← m
return A
```

- Running time: $\Theta(n)$ if activities are already sorted by finish time; else $\Theta(n\log n)$

Why is it Greedy?

- Greedy in the sense that it leaves as much opportunity as possible for the remaining activities to be scheduled.
- The greedy choice is the one that maximizes the amount of unscheduled time remaining.

- We will show that this algorithm uses the following properties
  - The algorithm satisfies the greedy-choice property
  - The problem has the optimal substructure property

Elements of Greedy Strategy

- A greedy algorithm makes a sequence of choices, each of the choices that seems best at the moment is chosen
  - NOT always produce an optimal solution
- Two ingredients that are exhibited by most problems that lend themselves to a greedy strategy
  - Greedy-choice property
  - Optimal substructure

Greedy-Choice Property

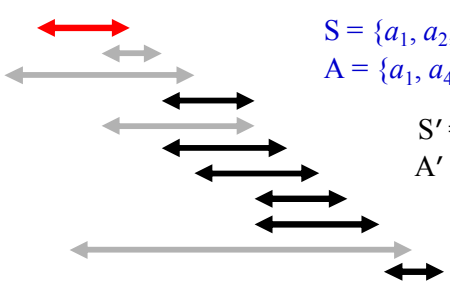
- A globally optimal solution can be arrived at by making a locally optimal (greedy) choice
  - Make whatever choice seems best at the moment and then solve the sub-problem arising after the choice is made
  - The choice made by a greedy algorithm may depend on choices so far, but it cannot depend on any future choices or on the solutions to sub-problems
- Of course, we must prove that a greedy choice at each step yields a globally optimal solution

Greedy-Choice Property

- Show there is an optimal solution that begins with a greedy choice (with activity 1, which as the earliest finish time)
- Suppose A $\subseteq$ S in an optimal solution
  - Order the activities in A by finish time. The first activity in A is k
    - If $k = 1$, the solution A begins with a greedy choice
    - If $k \neq 1$, then there is an optimal solution B to S that begins with the greedy choice, activity 1
  - Let B = A – $\{k\} \cup \{1\}$
    - $f_1 \leq f_k$ → activities in B are disjoint (compatible)
    - B has the same number of activities as A
    - Thus, B is optimal

Optimal Substructure Property

- A problem exhibits optimal substructure if an optimal solution to the problem contains within it optimal solutions to sub-problems
  - If an optimal solution A to S begins with activity 1, then A′ = A – {1} is optimal to S′ ={i ∈S: $s_i \geq f_1$}



S = $\{a_1, a_2, a_3, a_4, a_5, a_6, a_7, a_8, a_9, a_{10}, a_{11}\}$
A = $\{a_1, a_4, a_8, a_{11}\}$

S′ = $\{a_4, a_6, a_7, a_8, a_9, a_{11}\}$
A′ = $\{a_4, a_8, a_{11}\}$

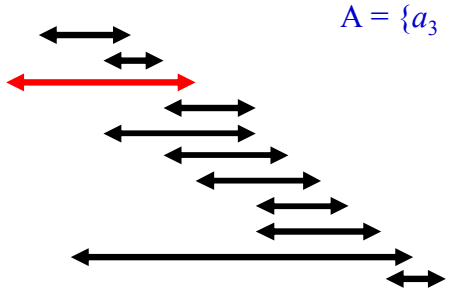
Optimal Substructure Property

- Once the greedy choice of activity 1 is made, the problem reduces to finding an optimal solution for the activity-selection problem over those activities in S that are compatible with activity 1
  - Optimal Substructure
  - If A is optimal to S, then A′ = A – {1} is optimal to $S' = \{i \in S: s_i \geq f_1\}$
  - Why?
    - If we could find a solution B′ to S′ with more activities than A′, adding activity 1 to B′ would yield a solution B to S with more activities than A ➔ contradicting the optimality of A
- After each greedy choice is made, we are left with an optimization problem of the same form as the original problem
  - By induction on the number of choices made, making the greedy choice at every step produces an optimal solution

Early Start Greedy

- Select the activity with the earliest start
- Eliminate the activities that could not be scheduled
- Repeat!

A = {a_3 }



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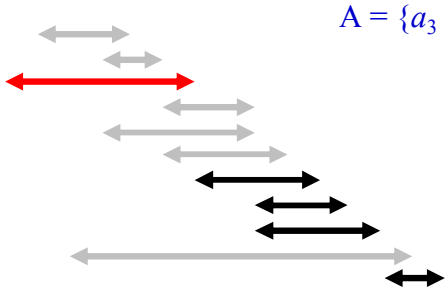
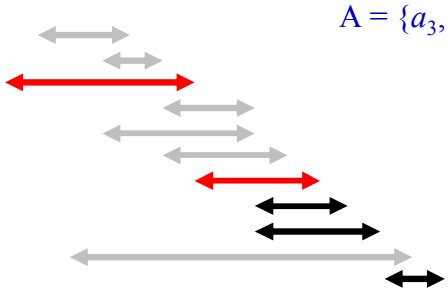


Early Start Greedy

- Select the activity with the earliest start
- Eliminate the activities that could not be scheduled
- Repeat!

A = {a_3, a_7 }



Early Start Greedy

- Select the activity with the earliest start
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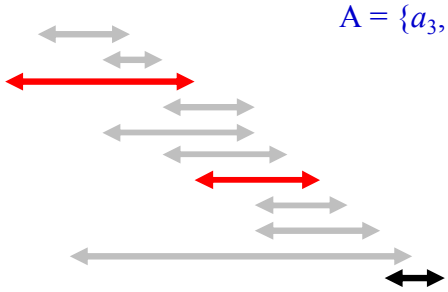
A = $\{a_3, a_7 \quad \}$

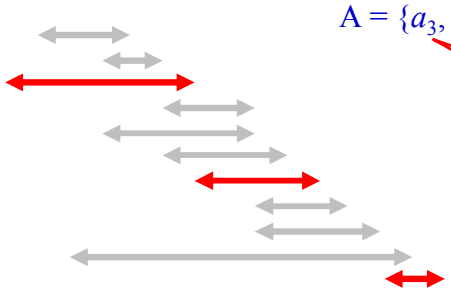


Early Start Greedy

- Select the activity with the earliest start
- Eliminate the activities that could not be scheduled
- Repeat!

A = $\{a_3, a_7, a_{11}\}$

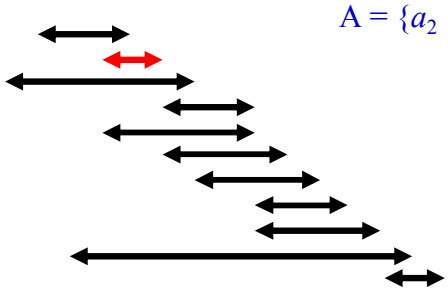
Not an optimal solution



Shortest Interval Greedy

- Select the activity with the *shortest interval*
- Eliminate the activities that could not be scheduled
- Repeat!

A = {a_2 }



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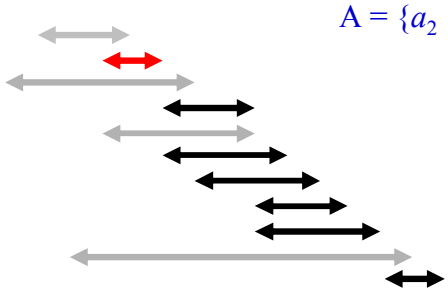
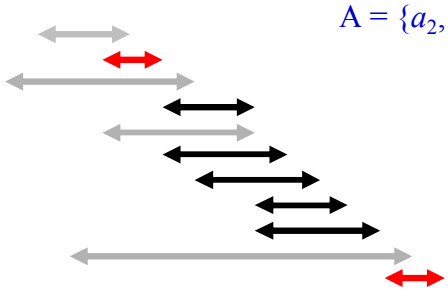


Shortest Interval Greedy

- Select the activity with the *shortest interval*
- Eliminate the activities that could not be scheduled
- Repeat!

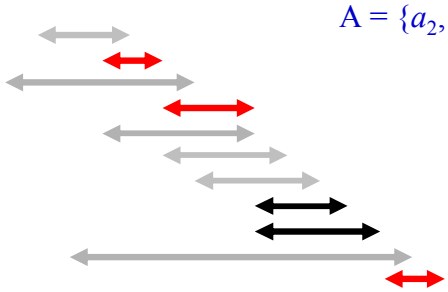
A = {a_2, a_{11} }



Shortest Interval Greedy

- Select the activity with the *shortest interval*
- Eliminate the activities that could not be scheduled
- Repeat!

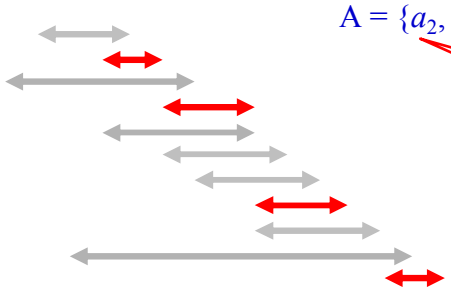
A = {a_2, a_{11}, a_4 }



Shortest Interval Greedy

- Select the activity with the *shortest interval*
- Eliminate the activities that could not be scheduled
- Repeat!

A = $\{a_2, a_{11}, a_4, a_8\}$

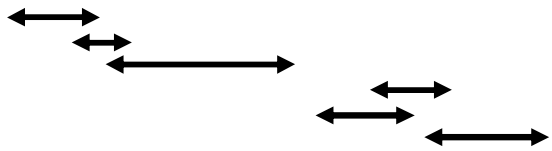


Gives optimal solution in this example

Shortest Interval Greedy

- Select the activity with the *shortest interval*
- Eliminate the activities that could not be scheduled
- Repeat!

A = { }



Shortest Interval Greedy

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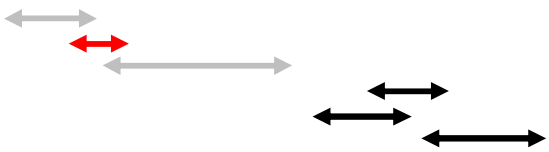
- Repeat!

A = {a_2 }

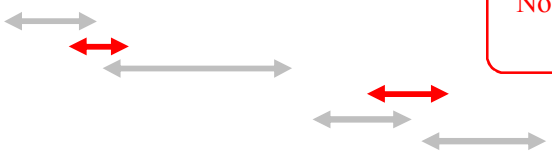


Shortest Interval Greedy

- Select the activity with the *shortest interval*
- Eliminate the activities that could not be scheduled
- Repeat!

A = {a_2, a_4}

Not an optimal solution



Shortest Interval Greedy

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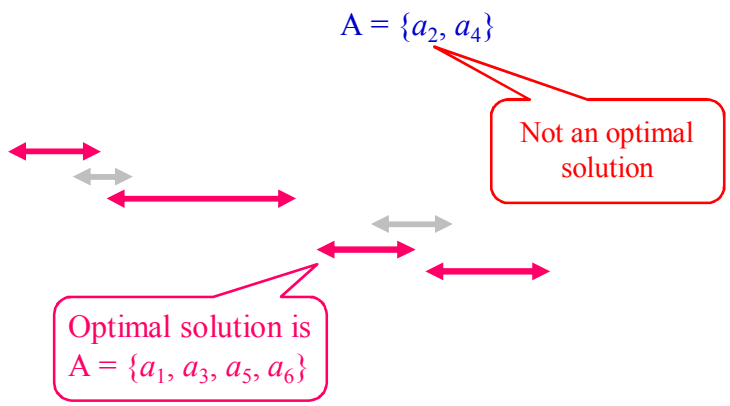

Sample Question

Suppose that you run a car lending business, and each request is at the format: [start time, end time). A car can be used by one user at a time. Suppose that you have got the following car-use requests for the next day.

{ [1, 6), [4, 8), [6, 8), [9, 15), [6, 9), [11, 15), [2, 6) }

Find *all solutions* of the maximum number of requests that can be satisfied by using

i) earliest end time,

ii) shortest interval, and

iii) earliest start time.

All Solutions for Earliest End Time

{ [1, 6), [4, 8), [6, 8), [9, 15), [6, 9), [11, 15), [2, 6) }

All Solutions for Shortest Interval

{ [1, 6), [4, 8), [6, 8), [9, 15), [6, 9), [11, 15), [2, 6) }

All Solutions for Earliest Start Time

{ [1, 6), [4, 8), [6, 8), [9, 15), [6, 9), [11, 15), [2, 6) }